Content-Centric Modelling of Rage-Bait Lifecycles on Social Media with Toxicity Feedback

APM348 — 10-Minute Presentation

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Why Rage-Bait?

Emotional, morally charged content spreads further and faster than neutral posts on every major platform [1]. Platforms know this, and they tune their algorithms to ride it. But what happens downstream?

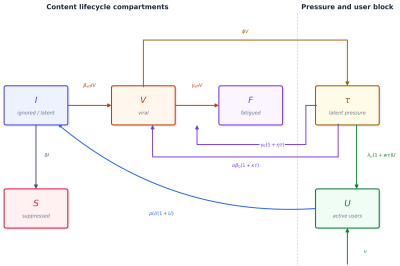
Rage-bait = content that gets people *arguing*, not just sharing. High replies relative to retweets $\Rightarrow$ high τ/V.

The Higgs boson discovery triggered exactly this: replies flooded in way beyond the retweet count. Science, not politics, same pattern.

Central question

How does algorithmic amplification shape the full lifecycle of rage-bait, and what does it cost in toxicity and user retention?

Model Structure



Deterministic compartmental ODE. $\beta_{\text{eff}} = \alpha\beta_0(1+\kappa\tau)$: latent pressure τ feeds back into spread, fatigue, and user retention.

Variable	Meaning
I	Ignored content (unseen)
V	Viral content (spreading)
F	Fatigued content (stale)
S	Suppressed content (removed)
τ	Discussion pressure (latent toxicity)
U	Active user pool

Key idea: we classify *content*, not users.

Content volumes are directly observable; user states are not.

Content Dynamics

$$\dot{I} = \frac{\rho\, U}{1+U} - \beta_{\text{eff}}\, I\, V - (\delta + \mu_c)\, I$$

$$\dot{V} = \beta_{\text{eff}}\, I\, V - \gamma_{\text{eff}}\, V - \mu_c\, V$$

$$\dot{F} = \gamma_{\text{eff}}\, V - \mu_c\, F$$

$$\dot{S} = \delta\, I - \mu_c\, S$$

Effective rates

$\beta_{\text{eff}} = \alpha\, \beta_0\, (1 + \kappa\, \tau)$ $\gamma_{\text{eff}} = \gamma_0\, (1 + \eta\, \tau)$

$\alpha =$ platform amplification strength. $\kappa =$ toxicity boost to spread. $\eta =$ toxicity-accelerated fatigue.

Pressure and User Dynamics

$$\dot{\tau} = \phi\, V - \psi\, \tau$$

$$\dot{U} = \nu - \lambda_u\,(1 + w\,\tau)\, U$$

Pressure equation

Viral content generates discussion pressure at rate ϕ.
Pressure decays naturally at rate ψ.

User equation

Users arrive at rate ν and leave at baseline rate λ_u.
Toxicity multiplies departure by $(1 + w\,\tau)$.

Feedback: $V \xrightarrow{\phi} \tau \xrightarrow{\kappa} \beta_{\text{eff}} \rightarrow V$ **:** a self-reinforcing toxicity loop.

Well-Posedness and the Basic Reproduction Number

Well-posedness

- **Nonnegativity:** every face of $\mathbb{R}^6_{\geq 0}$ is inward-pointing
- **Boundedness:** $U \leq \nu/\lambda_u, \quad C \leq \rho/\mu_c,$ $\tau \leq \phi\, C_{\max}/\psi$

Solutions exist globally and stay in the positive cone.

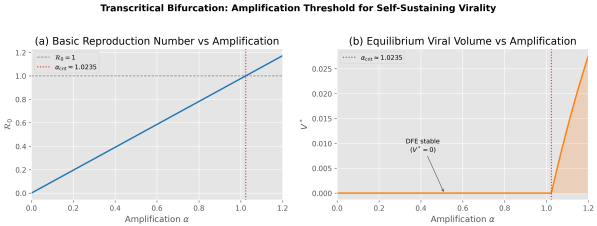
Amplification threshold

$$\mathcal{R}_0 = \frac{\alpha\, \beta_0\, I_{\mathrm{DFE}}}{\gamma_0 + \mu_c}$$

$$\boxed{\alpha_{\mathrm{crit}} \approx 0.3165}$$

$\mathcal{R}_0 < 1$: virality dies out (DFE stable).
$\mathcal{R}_0 > 1$: endemic virality persists.

Bifurcation and Stability



Transcritical bifurcation

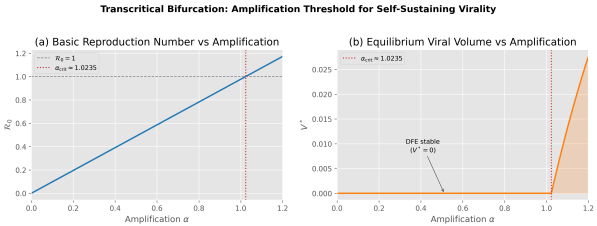
At $\alpha = \alpha_{\text{crit}}$, the disease-free and endemic equilibria exchange stability.

- DFE eigenvalue for V:

$$(\gamma_0 + \mu_c)(\mathcal{R}_0 - 1)$$

- Positive equilibrium: numerically verified LAS for all tested $\alpha > \alpha_{\text{crit}}$

The Toxicity Feedback Loop



The cycle

$$V \xrightarrow{\phi} \tau \xrightarrow{\kappa} \beta_{\text{eff}} \rightarrow V$$

1. Viral content drives discussion pressure (ϕ)
2. Pressure amplifies spread rate (κ)
3. Faster spread creates more viral content

Brakes: fatigue (γ_{eff}), pressure decay (ψ), and user departure ($w\tau$).

Calibration

Spread block ✓

- Higgs Twitter retweet cascade, 100 h window
- Full-window $R^2 = 0.92$
- Narrow bootstrap confidence intervals
- Beats SIR benchmark (spike $R^2 = 0.89$ vs. 0.69)

Pressure block ∼

- Reply activity as proxy for τ
- Anchored to external Jigsaw toxicity scale
- ϕ, ψ not uniquely identified
- Qualitative policy ranking is robust across all five calibration configs

Spread calibration is strong; pressure block is proxy-grounded.

Results: Three Amplification Scenarios

Scenario	α	$\mathcal{R}_0$	V^*	τ^*	U^*
Engagement-first	0.9	2.84	0.1203	0.0674	29.88
Moderate	0.5	1.58	0.0697	0.0390	35.97
Health-first	0.2	0.63	0.0000	0.0000	50.00

The cost of amplification

Going from moderate to aggressive:

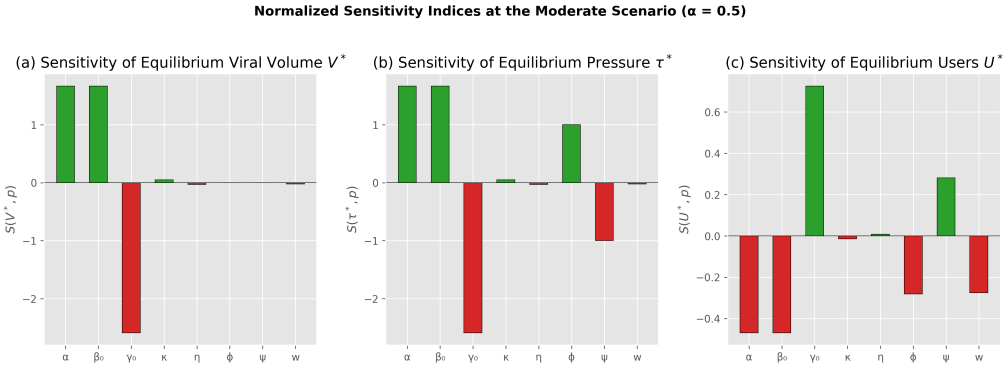
- Viral volume up **+73%**
- Discussion pressure up **+73%**
- User base down **−40%**

The escape

Below $\alpha_{\text{crit}} \approx 0.3165$:

- No sustained virality
- Zero endemic pressure
- Full user retention

Sensitivity Analysis



- Spread parameters $(\gamma_0, \alpha, \beta_0)$ dominate viral volume.
- Pressure parameters (ϕ, w) control user retention.
- Policy levers separate: content tuning $\neq$ retention tuning.

Conclusions

1. **Threshold.** A sharp amplification threshold exists ($\alpha_{\text{crit}} \approx 0.3165$). Below it, no sustained virality.
2. **Feedback loop.** Above threshold, a self-reinforcing cycle locks in: viral content $\rightarrow$ pressure $\rightarrow$ amplified spread $\rightarrow$ more viral content.
3. **Quantifiable tradeoff.** Aggressive amplification buys engagement today at the cost of a smaller, more toxic, less sustainable platform tomorrow.

Thank you. Questions?

Code and report: `github.com/n33levo/apm348_project`

Backup: Full ODE System

$$\dot{I} = \frac{\rho\, U}{1+U} - \beta_{\text{eff}}\, I\, V - (\delta + \mu_c)\, I$$

$$\dot{V} = \beta_{\text{eff}}\, I\, V - \gamma_{\text{eff}}\, V - \mu_c\, V$$

$$\dot{F} = \gamma_{\text{eff}}\, V - \mu_c\, F$$

$$\dot{S} = \delta\, I - \mu_c\, S$$

$$\dot{\tau} = \phi\, V - \psi\, \tau$$

$$\dot{U} = \nu - \lambda_u (1 + w\tau)\, U$$

$$\beta_{\text{eff}} = \alpha\, \beta_0\, (1 + \kappa\, \tau), \qquad \gamma_{\text{eff}} = \gamma_0\, (1 + \eta\, \tau)$$

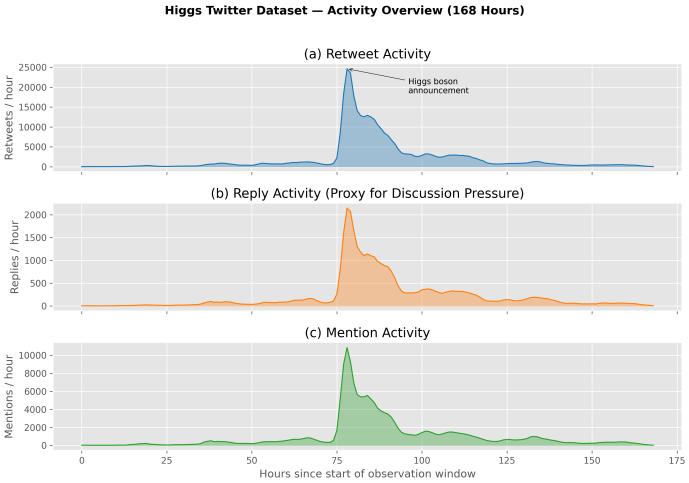
Backup: Pressure Configurations

Configuration	ϕ	ψ	Proxy R^2	Moderate τ^*
Current baseline	0.056	0.100	0.18	0.039
Higgs decay-anchored	0.061	0.041	0.10	0.104
External scale	0.146	0.100	0.31	0.102
Reference-constrained	0.417	0.286	0.41	0.102
Higgs (boundary-constrained)†	0.960	1.000	0.74	0.067

†Optimizer reached upper bound $\psi = 1.0$; true unconstrained optimum may have $\psi > 1$.

The qualitative policy ranking is preserved across all five configurations.

Backup: Higgs Dataset Overview



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